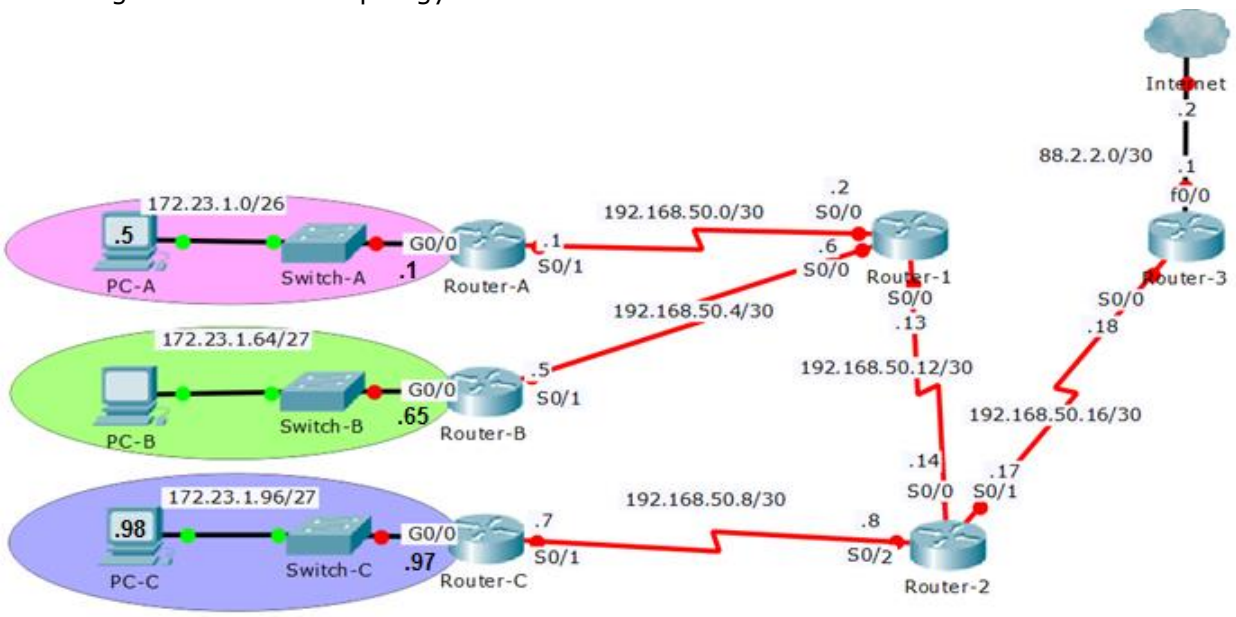
MCQ: Choose the correct answers

1) A network administrator has examined the routing table of a router and noted that the entry for the destination network 172.16.4.0/24 begins with the letter **D**.
   What does this letter signify?
   a. The route to network 172.16.4.0/24 is directly connected.
   b. The route source was learned dynamically.
   c. That is the direct route for packets to that network.
   d. The route to this network is configured statically on the router
2) What type of route has a network address with a subnet mask that is less than the classful equivalent?
   a. a network route
   b. a default route
   c. a supernet route
   d. a child route
3) Which two route source codes are automatically created within a routing table whenever a router interface is configured with an IP address and activated? (Choose two.)
   a. C
   b. O
   c. L
   d. R
   e. D
4) Choose the correct statements:
   a. A parent route is a level 1 network route that is subnetted.
   b. Level 2 child routes are also ultimate routes.
   c. Level 1 parent routes does not contain level 2 child routes.
   d. All of the above.
5) Which of the following is true regarding static routing?
   a. More secure.
   b. Uses CPU, memory and link bandwidth.
   c. Suitable for complex topologies.
   d. All of the above.
6) The administrative distance of the protocol
   a. identifies the trustworthiness of a route source
   b. identifies the value assigned to reach a remote network
   c. identifies how the route was learned
   d. identifies the address of a remote network

Routing Configuration (Static and RIP)

You are given the below topology



1) Configure the **static routes** entries on **Router-3** in order to have full network connectivity. **Minimize** the routing entries as much as it is convenient. (Use next hop address, instead of exit interface).

```
Router-3(config)#ip route 172.23.1.0 255.255.255.128 192.168.50.17
Router-3(config)#ip route 192.168.50.0 255.255.255.240 192.168.50.17
Router-3(config)#ip route 0.0.0.0 0.0.0.0 88.2.2.2
```

2) Configure the **static routes** entries on **Router-2** in order to have full network convergence. **DO NOT minimize** the number of route entries. (Use exit interface instead of next hop address).

```
Router-2(config)#ip route 172.23.1.0 255.255.255.192 S0/0
Router-2(config)#ip route 172.23.1.64 255.255.255.224 S0/0
Router-2(config)#ip route 192.168.50.0 255.255.255.252 S0/0
Router-2(config)#ip route 192.168.50.4 255.255.255.252 S0/0
Router-2(config)#ip route 172.23.1.96 255.255.255.224 S0/2
Router-2(config)#ip route 88.2.2.0 255.255.255.252 S0/1
Router-2(config)#ip route 0.0.0.0 0.0.0.0 S0/1
```

3) Answer the following questions:

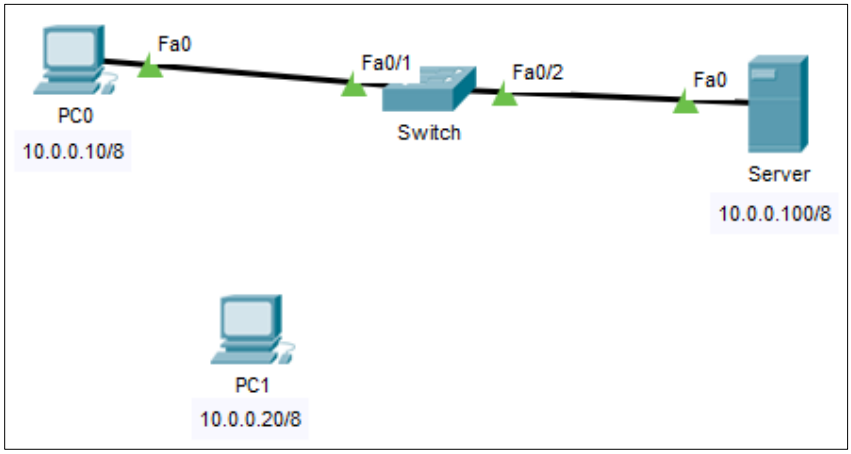
- Which command will advertise the default route in RIP updates? *default-information originate* (on Router-3)
- What is a "Floating static route" They are static routes that are used to provide a backup path to a primary static or dynamic route, in the event of a link failure
- Is it useful to define a Floating static default route on Router-3? No because it has only one path for the Internet

4) Enable and configure **RIPv2** for the networks directly connected to **Router-C**; disable the summarization of networks, and configure the LAN interfaces as passive.

Router rip
Version 2
Network 192.168.50.8
Network 172.23.1.96
No auto-summary
Passive interface g0/0

Switchport Port Security Configuration

Consider the following topology. Answer the below questions:



1. PC0 is connected with Fa0/1 port of the Switch.

a) Enable port security feature on this port	*Switch(config-if)#Switchport port security*
b) Set the maximum number of hosts that can be associated with this port to 1	*Switch(config-if)#Switchport port security maximum 1*
c) Enable the Sticky feature	*Switch(config-if)#Switchport port security mac-address sticky*

2. Switchport port security testing

In the previous topology, we have one additional PC (i.e. PC1). Assume that, this is the cracker's PC. To gain unauthorized access in network he unplugged the Ethernet cable from PC0 and plugged in his pc (PC1).
From PC1 try to ping the Server IP:

a) Did the ping command get response from the Server? ……No……
b) Why? Because the switch detected the mac address change
c) What does the switch do regarding its Fa0/1 port? It shut down this port (default violation mode)